

// Day 30 of 30

COMPLETE ROADMAP

Spring Boot Project Roadmap — 0 to Production

→ Swipe through all 7 slides



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Phase 1: Project Setup

Day 1 — Foundation

1. `start.spring.io`
 - Spring Web, JPA, MySQL, Security
 - Lombok, DevTools, Actuator
2. Project structure:
 - `controller/`
 - `service/`
 - `repository/`
 - `model/`
 - `dto/`
 - `config/`
 - `exception/`

💡 Good structure = good project. Always.



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Phase 2: Database + Entities

Day 2-3 — Data layer

```
@Entity
public class User {
    @Id @GeneratedValue
    private Long id;
    private String name;
    @Column(unique = true)
    private String email;
}

public interface UserRepo
    extends JpaRepository {
    Optional findByEmail(String e);
}
```

💡 Design entities carefully. Schema changes are expensive.



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Phase 3: REST APIs

Day 4-5 — Core endpoints

```
@RestController
@RequestMapping("/api/v1/users")
public class UserController {

    @GetMapping("/{id}")
    public UserDTO getUser(@PathVariable Long id)

    @PostMapping
    public UserDTO createUser(@Valid @RequestBody)

    @PutMapping("/{id}")
    public UserDTO updateUser( ... )

    @DeleteMapping("/{id}")
    public void deleteUser( ... )
}
```

💡 Always version APIs: /api/v1/. Always use DTOs.



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Phase 4: Security + Auth

Day 6-7 — Lock it down

1. Spring Security config
2. JWT token generation
3. JWT validation filter
4. BCrypt password encoding
5. Role-based access:
`@PreAuthorize("hasRole('ADMIN')")`
6. CORS configuration
7. Rate limiting

💡 Security is not optional. Add it from day 1.



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Phase 5: Testing

Day 8-9 — Prove it works

Unit tests:

- `@ExtendWith(MockitoExtension.class)`
- Mock dependencies
- Test service logic

Integration tests:

- `@SpringBootTest`
- Test full flow
- Real DB (H2 in-memory)

API tests:

- `@WebMvcTest`
- Test controllers
- MockMvc requests

💡 No tests = no confidence. Write them.



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Phase 6: Docker + Deploy

Day 10 — Ship it

1. Dockerfile
2. docker-compose.yml (app + DB)
3. Environment variables for secrets
4. GitHub Actions CI/CD:
 - Run tests
 - Build Docker image
 - Push to registry
 - Deploy to server
5. Spring Actuator for health checks
6. Logging with SLF4J

💡 If it's not deployed, it doesn't exist.



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Phase 7: Production Readiness

The final 10% that matters most

Checklist:

- ❑ Error handling (@ControllerAdvice)
- ❑ Input validation (@Valid)
- ❑ Pagination on list endpoints
- ❑ Database indexes on query columns
- ❑ API documentation (Swagger)
- ❑ Health endpoint (/actuator/health)
- ❑ Logging (request ID tracking)
- ❑ Monitoring (Prometheus + Grafana)
- ❑ README with setup instructions

💡 This checklist = what separates juniors from seniors



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done = true

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